
Overview for Data Science

Chapter 2

Learning outcomes

After completing this lesson you should be able to

- Describe what data science is and the role of data scientists.

- Differentiate data and information.

- Describe data processing life cycle

- Understand different data types from diverse perspectives

- Describe data value chain in emerging era of big data

- Basic concepts of Big Data

An Overview of Data Science

Data science is a multidisciplinary field that uses

scientific methods, processes and algorithm systems to extract knowledge, Insights from structured, semi-structured and unstructured data

Data science continues to evolve as one of the most promising and in-demand career paths for skilled professionals

What is data?

A representation of **facts**, **concepts**, or **instructions** in a formalized manner, which should be **suitable** for communication, interpretation, or **processing** by human or electronic machine

Data can be described as **unprocessed facts** and **figures**

It can also be defined as groups of **non-random symbols** in the form of **text**, **images**, and **voice** representing quantities, action and objects

What is Information?

Organized or classified data, which has some meaningful values for the receiver

Processed data on which decisions and actions are based. Plain collected data as raw facts cannot help much in decision-making

Interpreted data created from organized, structured, and processed data in a particular context

Data Processing Cycle

Data processing is the **re-structuring** or **re-ordering** of data by people or machine to increase their usefulness and add values for a particular purpose

Data processing consists the following steps

Input

Processing

Output



Data Processing Cycle

Input

The input data is prepared in some convenient form for processing

The form will depend on the processing machine

For example, when electronic computers are used, the input data can be recorded on any one of the several types of input medium, such as flash disks, hard disk, and so on

Data Processing Cycle

Processing

In this step, the input data is changed to produce data in a more useful form

For example, a summary of sales for a month can be calculated from the sales orders data

Data Processing Cycle

Output

At this stage, the result of the proceeding processing step is collected

The particular form of the output data depends on the use of the data

For example, output data can be total sale in a month

Data types and its representation

In computer science and computer programming, a **data type** or simply type is an attribute of data which tells the compiler or interpreter **how the programmer intends to use the data**

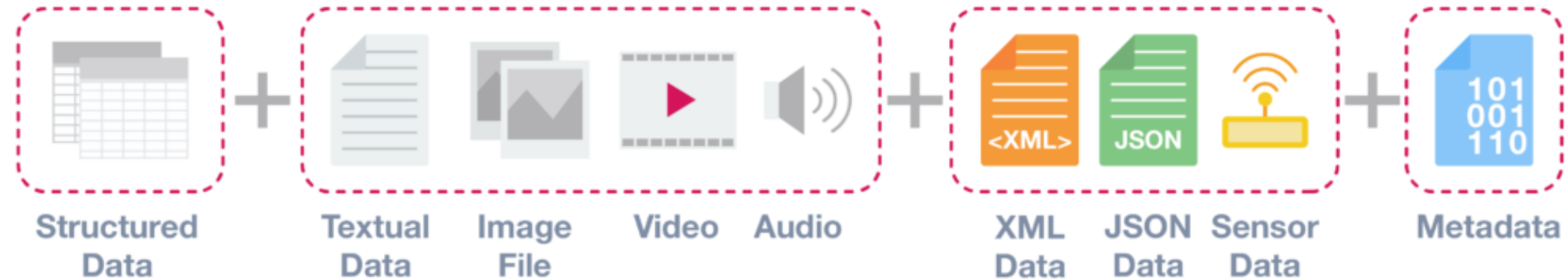
Common data types include

Integers, Boolean, Characters, Floating-Point Numbers, Alphanumeric Strings

This data type **defines the operations that can be done on the data, the meaning of the data, and the way values of that type can be stored**

Data types from Data Analytics perspective

Structured, Unstructured, and Semi-structured data types



Structured Data

Data that adheres to a **predefined data model** and is therefore straightforward to analyze

Conforms to a **tabular format** with relationship between different rows and columns

Common examples

Excel files or SQL databases

Unstructured Data

Data that either **does not** have a predefined data model or is not organized in a predefined manner

It is typically **text-heavy**, but may contain data such as **dates**, **numbers**, and **facts** as well

Common examples

audio, video files, NoSQL, pictures, pdfs ...

Semi-structured Data

A form of structured data that **does not conform with the formal structure of data models** associated with relational databases or other forms of data tables

But **contain tags** or other **markers to separate semantic elements** and enforce hierarchies of records and fields within the data

Therefore, it is also known as **self-describing structure**

Semi-structured Data

Examples of semi-structured data

JSON and XML

```
{ "employees": [  
  { "firstName": "John", "lastName": "Doe" },  
  { "firstName": "Anna", "lastName": "Smith" },  
  { "firstName": "Peter", "lastName": "Jones" }  
]}
```

```
<employees>  
  <employee>  
    <firstName>John</firstName> <lastName>Doe</lastName>  
  </employee>  
  <employee>  
    <firstName>Anna</firstName> <lastName>Smith</lastName>  
  </employee>  
  <employee>  
    <firstName>Peter</firstName> <lastName>Jones</lastName>  
  </employee>  
</employees>
```

Metadata – Data about Data

It provides additional information about a specific set of data

For example

Metadata of a photo could describe **when** and **where** the **photos were taken**

The metadata then provides fields for **dates** and **locations** which, by themselves, can be considered structured data

Data Value Chain

Describe the information flow within a big data system as a **series of steps** needed to **generate value** and **useful insights** from data

The **Big Data Value Chain** identifies the following key high-level activities

Data Acquisition, Data Analysis, Data Curation, Data Storage, Data Usage,

Data Value Chain

Data Acquisition

- Structured data
- Unstructured data
- Event processing
- Sensor networks
- Protocols
- Real-time
- Data streams
- Multimodality

Data Analysis

- Stream mining
- Semantic analysis
- Machine learning
- Information extraction
- Linked Data
- Data discovery
- 'Whole world' semantics
- Ecosystems
- Community data analysis
- Cross-sectorial data analysis

Data Curation

- Data Quality
- Trust / Provenance
- Annotation
- Data validation
- Human-Data Interaction
- Top-down/Bottom-up
- Community / Crowd
- Human Computation
- Curation at scale
- Incentivisation
- Automation
- Interoperability

Data Storage

- In-Memory DBs
- NoSQL DBs
- NewSQL DBs
- Cloud storage
- Query Interfaces
- Scalability and Performance
- Data Models
- Consistency, Availability, Partition-tolerance
- Security and Privacy
- Standardization

Data Usage

- Decision support
- Prediction
- In-use analytics
- Simulation
- Exploration
- Visualisation
- Modeling
- Control
- Domain-specific usage

Data Acquisition

It is the process of **gathering**, **filtering**, and **cleaning** data before it is put in a data warehouse or any other storage solution on which data analysis can be carried out

Data acquisition is one of the major **big data challenges** in terms of **infrastructure requirements**

Data Acquisition

The infrastructure required for data acquisition must

- deliver **low, predictable latency** in both capturing data and in executing queries

- be able to **handle very high transaction volumes**, often in a distributed environment

- support **flexible and dynamic data structures**

Data Analysis

Involves **exploring**, **transforming**, and **modelling** data with the goal of **highlighting** relevant data, **synthesising** and **extracting** useful hidden information with high potential from a business point of view

Related areas include **data mining**, **business intelligence**, and **machine learning**

Data Curation

Active management of data over its life cycle to ensure it **meets** the necessary data quality requirements for its effective usage

Data curation processes can be categorized into different activities

content creation, selection, classification, transformation, validation, and preservation

Data Curation

Data curators (also known as scientific curators, or data annotators) hold the responsibility of ensuring that data are **trustworthy, discoverable, accessible, reusable**, and **fit** their purpose

A key trend for the curation of big data **utilizes community** and **crowd sourcing** approaches

Data Storage

It is the **persistence** and management of data in a scalable way that satisfies the needs of applications that require fast access to the data

Relational Database Management Systems (RDBMS) have been the main, and almost unique, solution to the storage paradigm for nearly 40 years

Data Storage

Relational database that guarantee database transactions, **lack flexibility** with regard to **schema changes**, **performance** and **fault tolerance** when data volumes and complexity grow, making them **unsuitable for big data scenarios**

NoSQL technologies have been designed with the **scalability** goal in mind and present a wide range of solutions based on alternative data models

Data Usage

Covers the **data-driven business activities** that need **access** to data, its **analysis**, and the tools needed to **integrate** the data analysis within the business activity

In business decision-making , it can **enhance competitiveness** through **reduction of costs, increased added value**, or any other parameter that can be measured against existing performance criteria

What Is Big Data?

A collection of data sets **so large and complex** that it becomes difficult to process using on-hand database management tools or traditional data processing applications

Big data is characterized by 3V and more

The Vs

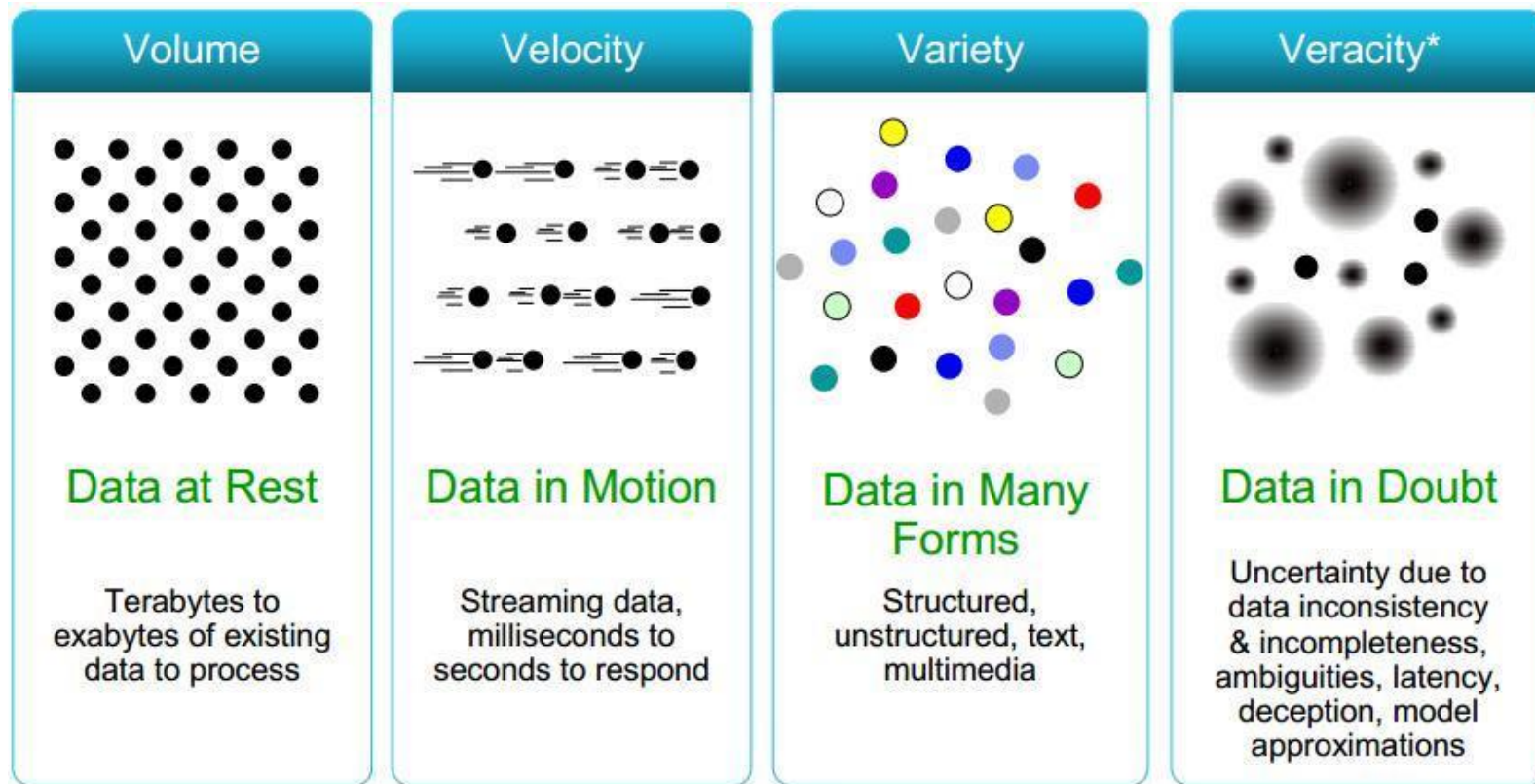
Volume: large amounts of data Zeta bytes/Massive datasets

Velocity: Data is live streaming or in motion

Variety: data comes in many different forms from diverse sources

Veracity: can we trust the data? How accurate is it? etc

The Vs



Clustered Computing

- Because of the qualities of big data, individual computers are often inadequate for handling the data at most stages.
 - To better address the high storage and computational needs of big data, computer clusters are a better fit
- Benefits :
 - **Resource Pooling:** Combining the available storage space to hold data is a clear benefit, but CPU and memory pooling are also extremely important. Processing large datasets requires large amounts of all three of these resources.

- **High Availability:** Clusters can provide varying levels of fault tolerance and availability guarantees to prevent hardware or software failures from affecting access to data and processing. This becomes increasingly important as we continue to emphasize the importance of real-time analytics.
- **Easy Scalability:** Clusters make it easy to scale horizontally by adding additional machines to the group. This means the system can react to changes in resource requirements without expanding the physical resources on a machine.